CO4 AT3 – Comparative Analysis
Solutions for Questions 14, 6, 3, 11 and 4

Question 14: Depth Estimation Error (Squared)

Given:

Object 1: Predicted by A = 2.5 m, B = 2.2 m, Actual = 2.0 m.

Object 2: Predicted by A = 3.5 m, B = 3.1 m, Actual = 3.0 m.

Object 3: Predicted by A = 4.5 m, B = 4.2 m, Actual = 4.0 m.

(a) Squared error = $(\text{Predicted depth} - \text{Actual depth})^2$.

Method A: Object 1 = $(2.5 - 2)^2 = 0.25\ m^2$; Object 2 = $(3.5 - 3)^2 = 0.25\ m^2$; Object 3 = $(4.5 - 4)^2 = 0.25\ m^2$.

Method B: Object 1 = $(2.2 - 2)^2 = 0.04\ m^2$; Object 2 = $(3.1 - 3)^2 = 0.01\ m^2$; Object 3 = $(4.2 - 4)^2 = 0.04\ m^2$.

Average squared error: A = $(0.25 + 0.25 + 0.25)/3 = 0.25\ m^2$. B = $(0.04 + 0.01 + 0.04)/3 = 0.03\ m^2$.

(b) Method B is more accurate because it has the lower average squared error ($0.03\ m^2$ compared with $0.25\ m^2$ for Method A).

Question 6: Precision–Recall Trade-off Analysis

Given: Model A: TP = 70, FP = 30, FN = 20. Model B: TP = 85, FP = 25, FN = 25.

Precision = TP / (TP + FP). Recall = TP / (TP + FN).

Model A: Precision = 70/(70+30) = 0.70 = 70%. Recall = 70/(70+20) = 0.7778 ≈ 77.78%.

Model B: Precision = 85/(85+25) = 0.7727 ≈ 77.27%. Recall = 85/(85+25) = 0.7727 ≈ 77.27%.

Evaluation: Model A has slightly higher recall, while Model B has higher precision. Model B is more balanced because its precision and recall are equal (both ≈77.27%).

Question 3: Optical Flow Magnitude Comparison

Given motion magnitudes: R1: A = 5, B = 8; R2: A = 6, B = 10; R3: A = 5, B = 9 pixels/frame.

(a) Average magnitude for Method A = (5 + 6 + 5)/3 = 16/3 = 5.33 pixels/frame.

Average magnitude for Method B = (8 + 10 + 9)/3 = 27/3 = 9.00 pixels/frame.

(b) Method B captures stronger motion because its average optical-flow magnitude (9.00 pixels/frame) is greater than Method A's (5.33 pixels/frame).

Question 11: Motion Estimation Variance

Given errors: Frame 1: A = 2, B = 6; Frame 2: A = 3, B = 7; Frame 3: A = 2, B = 5.

(a) Mean error: Method A = (2 + 3 + 2)/3 = 7/3 = 2.33 pixels. Method B = (6 + 7 + 5)/3 = 18/3 = 6.00 pixels.

(b) Using population variance, Variance = $\Sigma(x - \text{mean})^2 / N$.

Method A: deviations are −0.33, 0.67, −0.33. Variance ≈ (0.11 + 0.44 + 0.11)/3 = 0.22 pixels^2.

Method B: deviations are 0, 1, −1. Variance = (0 + 1 + 1)/3 = 0.67 pixels^2.

Evaluation: Method A is more stable because it has lower variance (≈0.22 pixels^2 versus ≈0.67 pixels^2) and also has a lower mean error.

Question 4: Depth Estimation Comparison (Stereo vs SfM)

Actual depths: A = 2 m, B = 3 m, C = 4 m.

Stereo estimates: 2.0, 3.0, 4.0 m. SfM estimates: 2.5, 3.8, 5.0 m.

(a) Absolute error = |Estimated depth − Actual depth|.

Stereo: A = |2.0−2| = 0 m; B = |3.0−3| = 0 m; C = |4.0−4| = 0 m.

SfM: A = |2.5−2| = 0.5 m; B = |3.8−3| = 0.8 m; C = |5.0−4| = 1.0 m.

Average absolute error: Stereo = (0 + 0 + 0)/3 = 0 m. SfM = (0.5 + 0.8 + 1.0)/3 = 2.3/3 ≈ 0.77 m.

(b) Stereo Vision is more accurate for the given data because its average absolute error is 0 m, compared with approximately 0.77 m for SfM.

Conclusion: Based on the calculated measures, Method B is preferable for Questions 14 and 6 where its error/precision-recall balance is better; Method B captures stronger optical flow in Question 3; Method A is more stable in Question 11; and Stereo Vision is more accurate in Question 4.